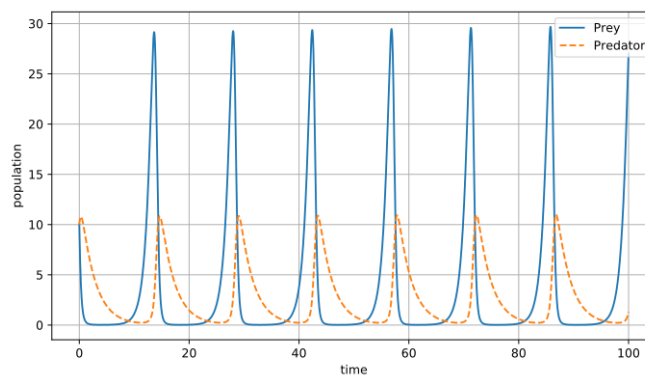


The Causes of Scarcity

Food scarcity is common among most animals, at least periodically. The causes of scarcity are multiple and often interrelated. Among animals that depends on plants as food, changes in climate that affect plant growth may result in long-term changes (positive or negative) to that food supply. In the more immediate setting, however, it is not long-term climate change but weather events, often described as natural hazards, that reduce food availability. A flood might destroy a wetland ecosystem in which a particular food plant grew; fire might burn a patch of forest, destroying most plants and removing all food from that area. Excessive cold or heat, or extreme drought, might kill even long-lived perennial plants.

The availability of plants that animals depend on as food supplies will also affect the populations of those animals. If there are many more animals who depend on a specific plant resource than that resource can support in a particular year, animal populations will be reduced. In some cases, this is due to starvation, while in others, the lack of resources means that females may not be fertile or will produce fewer young than usual, or that those young will be less likely to survive their first year, all of which reduce this animal's population in future years. The reduced population of this animal will then have an impact on predators that depend on the animal, leading to a set of complex interactions within an ecosystem as variations in the availability of specific resources result in changes in the populations of multiple animal species within a food web. Then, as a predator's population falls, that allows its prey's populations to recover. This relationship often takes the form of "boom and bust" cycles of both animals, which, in biology, are often defined simply by the Lotka-Volterra equation.

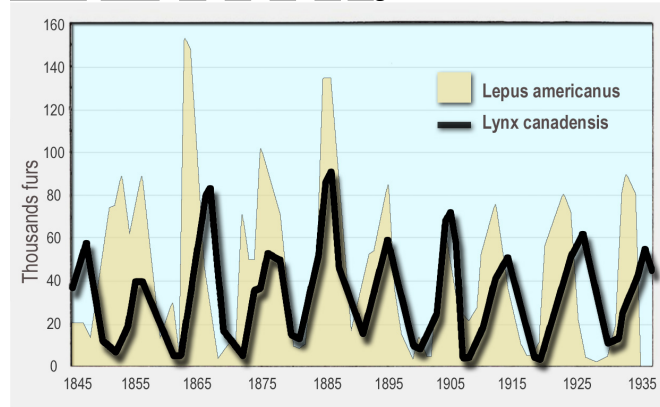


A simple predator-prey model created using the Lotka-Volterra equation.

Source: [Wikimedia Commons](#)

([https://commons.wikimedia.org/wiki/File:Lotka-](https://commons.wikimedia.org/wiki/File:Lotka-Volterra_model_(1.1,_0.4,_0.4,_0.1).svg)

[Volterra_model_\(1.1,_0.4,_0.4,_0.1\).svg](#))



A real-world dataset of predators (Canadian lynx, *Lynx canadensis*) and prey (snowshoe hares, *Lepus americanus*), as recorded by the number of furs of each sold to the Hudson Bay Company—a relationship that is generalized by the Lotka-Volterra equation.

Source: [Wikimedia Commons](#)

([https://commons.wikimedia.org/wiki/File:Milliers_fourrures_vendu](https://commons.wikimedia.org/wiki/File:Milliers_fourrures_vendues_environ_90_ans_odum_1953_en.jpg)

[es_en_environ_90_ans_odum_1953_en.jpg](#))

Environmental hazards are equally a threat to humans, especially those who practice agriculture. Farmers are limited by the areas that they have planted. If a farmer's field is hit by a tornado or hailstorm that destroys the crop just before harvest, which destroys the crop, then the farmer has nothing despite having spent a year planting and tending that crop. Farmers who depend on perennial crops, such as fruit trees, face even more challenging conditions, since it can take many years for trees to regrow and reach full fruit production. If trees are killed by fire or drought, it may take decades to recover.

Humans face a second set of threats to stable food supplies: other humans. War, raids, or even nonviolent expansion of other human groups into one's territory can diminish food supplies and result in scarcity. Historical accounts of wars going back to the earliest Mesopotamian texts record burning an enemy's fields as an established practice to introduce food scarcity and reduce an enemy's power; long sieges of hostile cities often followed this same strategy, starving an enemy until they surrendered. Of course, not all human conflict that leads to resource scarcity is overtly hostile. Instead, simple overpopulation in an area can lead to food scarcity, as can the sudden arrival of a large new population to a region, perhaps due to food scarcity (caused by environmental factors or human conflict) in their original home. Even today, large refugee populations can strain local food supplies, as we have seen resulting from conflicts such as the Syrian war.